Advanced Classical Mechanics

Lecture 10: Oscillations

Readings: Morin - Chapter 4, Marion Chapter 3

Amir Shahmoradi
Department of Physics / College of Science
The University of Texas at Arlington
Arlington, Texas

Hooke's Law

Consider an absolutely continuous *restoring* force (that tends toward an equilibrium position) whose Taylor expansion in terms of displacement can be written as,

$$F(x) = F_0 + x\left(\frac{dF}{dx}\right)_0 + \frac{1}{2!}x^2\left(\frac{d^2F}{dx^2}\right)_0 + \frac{1}{3!}x^3\left(\frac{d^3F}{dx^3}\right)_0 + \cdots$$

If the displacement x is defined with respect to a equilibrium position x_0, then F_0 must vanish at the equilibrium, leaving only $F(x) \approx -kx$ as the first-order approximation to $F(x)$ for small displacements x from equilibrium. This approximate equation is famously known as the Hooke's Law where $k \equiv -\left(\frac{dF}{dx}\right)_0$. Elastic deformations are an example class of processes that obey the Hooke's Law.

In reality, most oscillations in nature involve friction, which leads to a non-perfect **damped oscillation**. Systems experiencing an external force to keep the oscillations going are called **driven** (or **forced**) oscillations.

Simple Harmonic Motion (Ideal (Free) Oscillation: $\ddot{x} + \omega^2 x = 0$)

A system whose dynamics is described by the Hooke's Law is called a Harmonic Oscillator. The equation of motion can be written as $F = m\ddot{x} = -kx$. Defining $\omega_0^2 = \frac{k}{m}$ as the **angular frequency** of the system, the equation of motion can be written as $\ddot{x} + \omega_0^2 x = 0$. The solution to this homogeneous ODE is, $x(t) = A\sin(\omega_0 t - \delta)$ or $x(t) = A\cos(\omega_0 t - \phi)$ where A is the oscillation amplitude. The **period** of a Harmonic Oscillator is defined as the time interval between **successive repetitions of the particle's position and direction of motion**: $\omega_0\tau_0 = 2\pi \Rightarrow \tau_0 = 2\pi\sqrt{\frac{m}{k}}$.

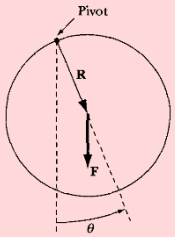
Problem 1. (Total energy of Harmonic Oscillator) Show that the total energy of a Harmonic oscillator is $\frac{1}{2}kA^2$. Is this total energy conserved?

Such systems as the simple Harmonic Oscillator whose period are independent of the amplitude (or energy) are called **isochronous** (meaning "occurring at the same time").

Simple and Damped Harmonic Motion

Problem 2. (Harmonic Pendulum)
Under what conditions a pendulum behaves like a simple harmonic oscillator? Derive the relevant equation of motion.

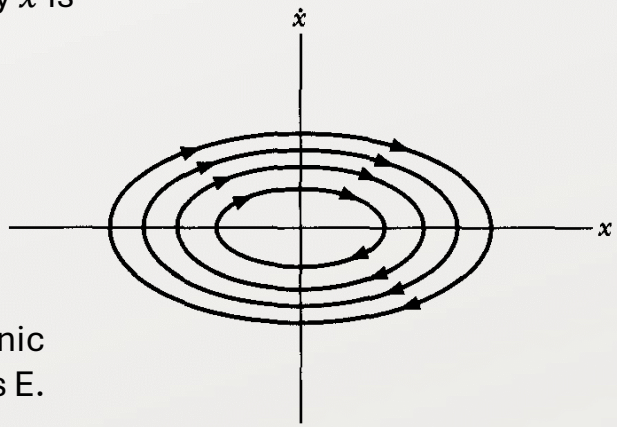
Problem 3. (Spherical Pendulum)
Find the angular velocity and period of small oscillation of a solid sphere of mass m and radius R about a point on its surface.



Phase Diagrams

The state of motion of a one-dimensional oscillator can be completely specified as a function of time if two quantities are given at one instant of time, that is, **the initial conditions $\boldsymbol{x(t_0)}$ and $\boldsymbol{\dot{x}(t_0)}$**. Two quantities are needed because the differential equation for the motion is of second order.) We may consider the quantities $\boldsymbol{x(t_0)}$ **and $\boldsymbol{\dot{x}(t_0)}$** to be the coordinates of a point in a two-dimensional space, called **phase space**. (In two dimensions, the phase space is a **phase plane**. But for a general oscillator with n degrees of freedom (n-dimensional oscillator), the phase space is a $2n$-dimensional space.) As the time varies, the point $P(x, \dot{x})$ describing the state of the oscillating particle will move along a certain phase path in the phase plane. For different initial conditions of the oscillator, the motion will be described by different phase paths. *Any given path represents the complete time history of the oscillator for a certain set of initial conditions*. The totality of all possible phase paths constitutes the **phase portrait** or the **phase diagram** of the oscillator.

If the coordinate axes of the phase plane are chosen as in the figure, the motion of the representative point $P(x, \dot{x})$ will always be in a clockwise direction, because for $x > 0$ the velocity $\dot{x}$ is always decreasing and for $x < 0$ the velocity is always increasing.



Phase diagram for a simple harmonic oscillator for a variety of total energies E.

Problem 4. (Phase Path Behavior) Can two phase paths cross each other? Justify your answer.

Problem 5. (Simple Harmonic Oscillator's Phase Path) Show that the phase path equation of a one-dimensional simple harmonic oscillator is an ellipse whose *squared* semi-major and semi-minor axes are $\frac{2E}{k}$ and $\frac{2E}{m}$ where E represents the total energy of the oscillator.

Damped Oscillations (More Realistic Oscillation: $\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = 0$)

The motion represented by the simple harmonic oscillator is termed a **free oscillation**; once set into oscillation, the **motion would never cease**. This oversimplifies the actual physical case, in which dissipative or frictional forces would eventually damp the motion to the point that the oscillations would no longer occur. We can analyze the motion in such a case by incorporating into the differential equation a term representing the damping force. It does not seem reasonable that the damping force should, in general, depend on the displacement, but it could be a function of the velocity or perhaps of some higher time derivative of the displacement. It is frequently **assumed that the damping force is a linear function of the velocity**, $F_d = av$. Since we consider here only one-dimensional damped oscillations so that we can represent the damping term by $-b\dot{x}$. The parameter b must be positive in order that the force indeed be resisting. (A force $-b\dot{x}$ with $b < 0$ would act to increase the speed instead of decreasing it as any resisting force must.) Thus, if a particle of mass m moves under the combined influence of a linear restoring force $-kx$ and a resisting force $-b\dot{x}$, the differential equation describing the motion is $m\ddot{x} + b\dot{x} + kx = 0$ which we will rewrite as, $\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = 0$ with $\beta \equiv \frac{b}{2m}$ knowns as the **damping parameter** and $\omega_0 = \sqrt{\frac{k}{m}}$ known as the **characteristic angular frequency** (in the absence of damping). The solution to this differential equation is,

$$x(t) = e^{-\beta t}\left[A_1 \exp\left(\sqrt{\beta^2 - \omega_0^2}\,t\right) + A_2 \exp\left(-\sqrt{\beta^2 - \omega_0^2}\,t\right)\right]$$

Three scenarios are possible:

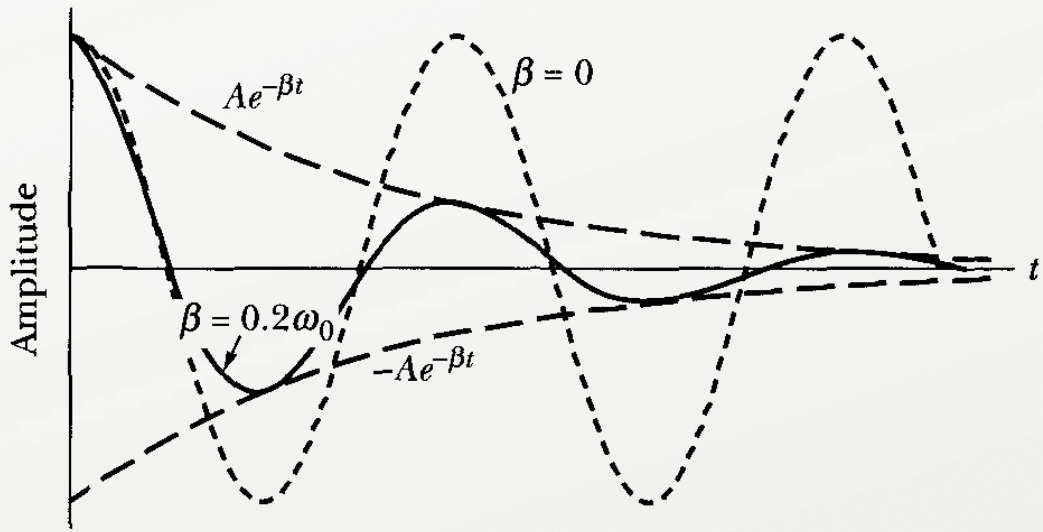
$$\begin{cases} \text{Underdamping: } \omega_0^2 > \beta^2 \\ \text{Critical damping: } \omega_0^2 = \beta^2 \\ \text{Overdamping: } \omega_0^2 < \beta^2 \end{cases}$$

Underdamped motion

For the case of underdamped harmonic oscillator, the equation of motion can be written as,

$$x(t) = e^{-\beta t}[A_1 \exp(i\omega_1 t) + A_2 \exp(-i\omega_1 t)] = Ae^{-\beta t}\cos(\omega_1 t - \delta)$$

where $\omega_1^2 = \omega_0^2 - \beta^2 > 0$ is the **angular frequency of the damped oscillator**. Strictly speaking, we cannot define a frequency when damping is present, because the motion is not periodic-that is, the oscillator never passes twice through a given point with the same velocity. However, because $w_1 = \frac{2\pi}{2T_1}$, where T_1 is the time between adjacent zero x-axis crossings, the angular frequency w_1 has meaning for a given time period. Note that $2T_1$ would be the "period" in this case, not T_1. For simplicity, we refer to w_1 as the "angular frequency" of the damped oscillator, and we note that this quantity is less than the frequency of the oscillator in the absence of damping (i.e., $w_1 < w_0$).



The underdamped motion (solid line) is an oscillatory motion (short dashes) that decreases within the exponential envelope (long dashes).

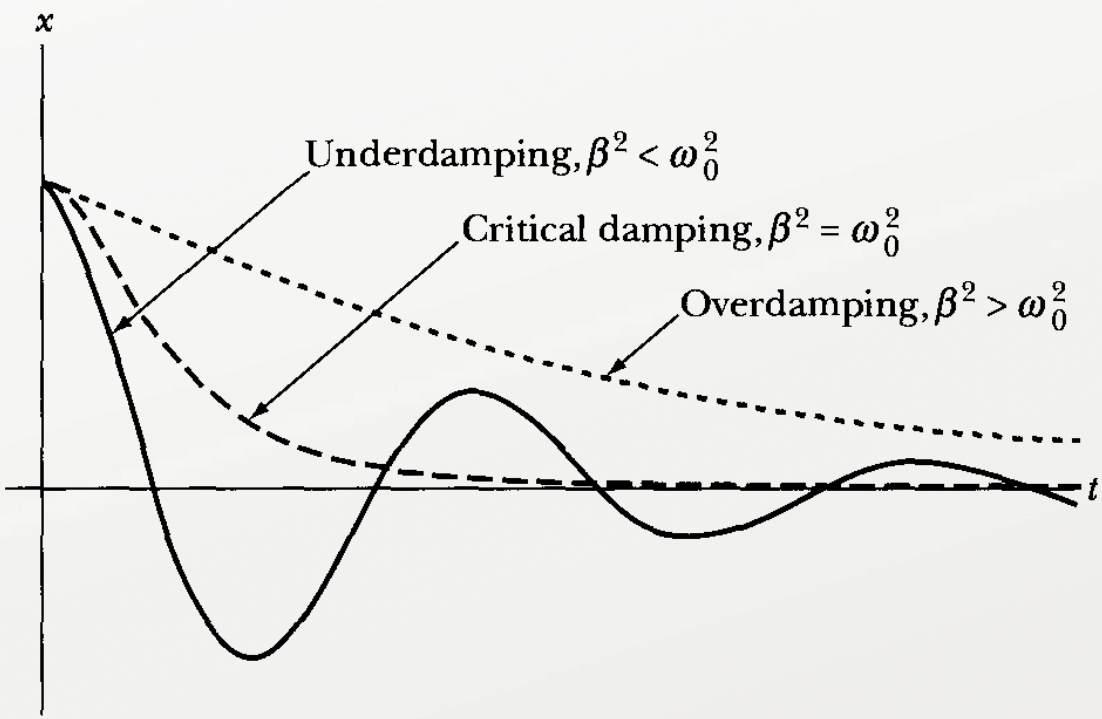
Problem 6. (Total energy of a Damped Harmonic Oscillator) Is the total energy of a damped harmonic oscillator constant over time? Derive an equation for the total energy of an underdamped oscillator.

Problem 7. (Phase Diagram of a Damped Harmonic Oscillator) Derive an equation for the phase diagram of an underdamped harmonic oscillator and depict its general behavior (for example, by assuming $A = 1\text{cm}$, $w_0 = 1\ \text{rad/s}$, $\beta = 0.2\text{s}^{-1}$, and $\delta = \frac{\pi}{2}\text{rad}$).

Critically Damped Motion

If the damping force is sufficiently large (i.e., if $\beta^2 > \omega_0^2$), the system is prevented from undergoing oscillatory motion. If zero initial velocity occurs, the displacement decreases monotonically from its initial value to the equilibrium position ($x = 0$). The case of critical damping occurs when β^2 is just equal to ω_0^2. The roots of the auxiliary equation are then equal, and the function x must be written as,

$$x(t) = (a + Bt)e^{-\beta t}$$



For a given set of initial conditions, a critically damped oscillator will approach equilibrium at a rate more rapid than that for either an overdamped or an underdamped oscillator. This is important in designing certain practical oscillatory systems (e.g., galvanometers) when the system must return to equilibrium as rapidly as possible. A pneumatic-tube screen-door closure system is a good example of a device that should be critically damped. If the closure were underdamped, the door would slam shut as other doors with springs always seem to do. If it were overdamped, it might take an unreasonably long time to close.

Overdamped Motion

If the damping parameter β is even larger than w_0 , then overdamping results. Because$\beta^2 > \omega_0^2$, the exponents in the brackets of the equation of motion become real quantities:

$$x(t) = e^{-\beta t}\left[A_1 \exp\left(\sqrt{\beta^2 - \omega_0^2}t\right) + A_2 \exp\left(-\sqrt{\beta^2 - \omega_0^2}t\right)\right] = e^{-\beta t}[A_1 \exp(\omega_2 t) + A_2 \exp(-\omega_2 t)]$$

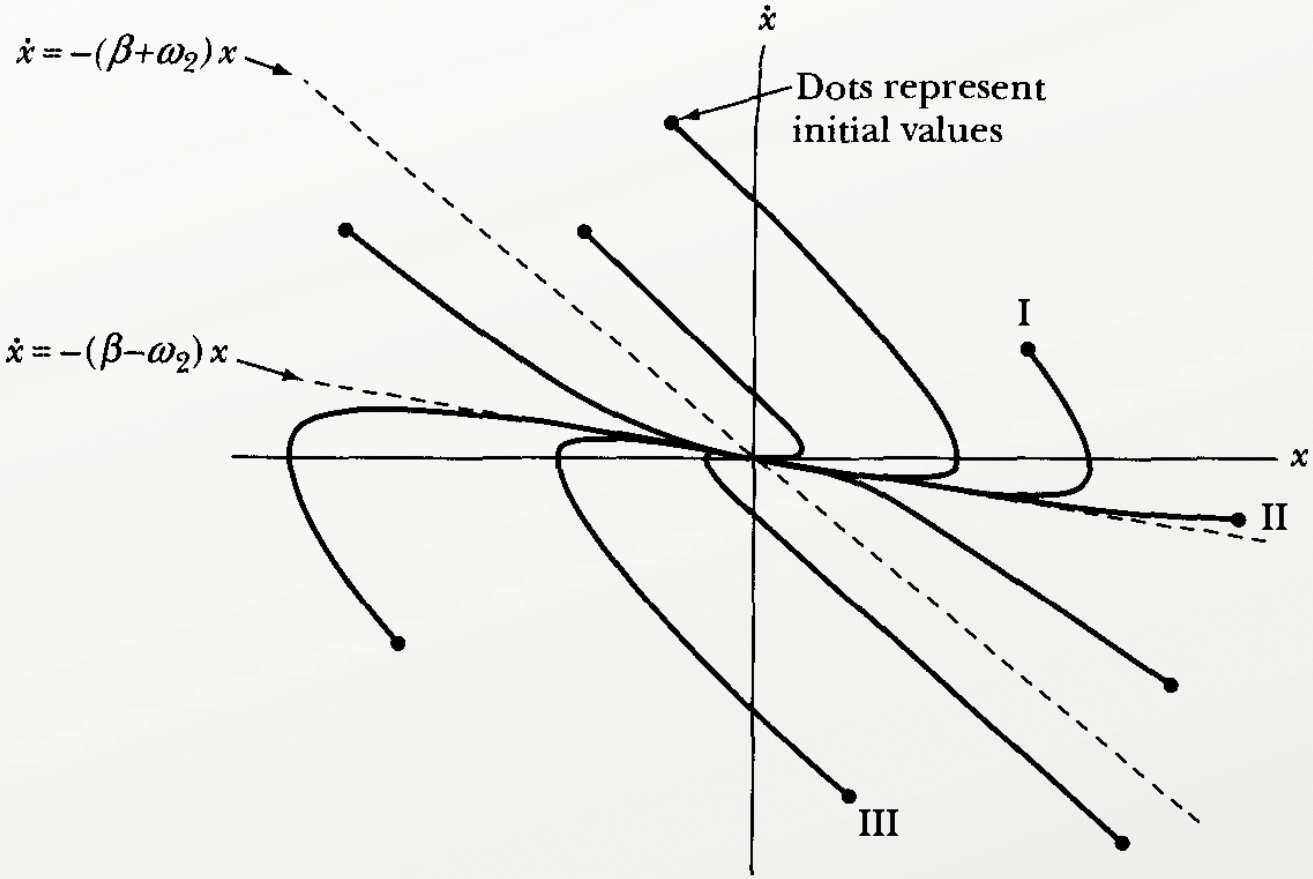
Note that w_2 does not represent an angular frequency, because the motion is not periodic. The displacement asymptotically approaches the equilibrium position. Overdamping results in a decrease of the amplitude to zero that may have some strange behavior as shown in the phase space diagram below. Notice that for all the phase paths of the initial positions shown, the asymptotic paths at longer times are along the dashed curve $\dot{x} = -(\beta - \omega_2)x$. Only a special case has a phase path along the other dashed curve. Depending on the initial values of the position and the velocity, a change in sign of both x and $\dot{x}$ may occur; for example, see the phase path labeled III in the figure. The figure on the next slide displays x and $\dot{x}$ as a function of time for the three phase paths labeled I, II, and III in the phase diagram. All three cases have initial positive displacements, $x(t = 0) = \mathrm{x}_0 > 0$. Each of the three phase paths has interesting behavior depending on the initial velocity value:

I. Case $\dot{x} > 0$, so that $x(t)$ reaches a maximum at some $t > 0$ before approaching zero. The velocity x decreases, becomes negative, and then approaches zero.
II. Case $\dot{x} < 0$, with $x(t)$ and $\dot{x}(t)$ reaching zero.
III. Case $\dot{x} < 0$, but below the curve $\dot{x} = -(\beta + w_2)\, x$, so that $x(t)$ goes negative before approaching zero, and $\dot{x}(t)$ goes positive before approaching zero. The motion in this case could be considered oscillatory.

The initial points lying between the two dashed curves in the phase diagram figure seem to have phase paths decreasing monotonically to zero, whereas those lying outside those two lines do not. Critical damping has phase paths similar to the overdamping curves shown in below, rather than the spiral paths of the underdamped motion.

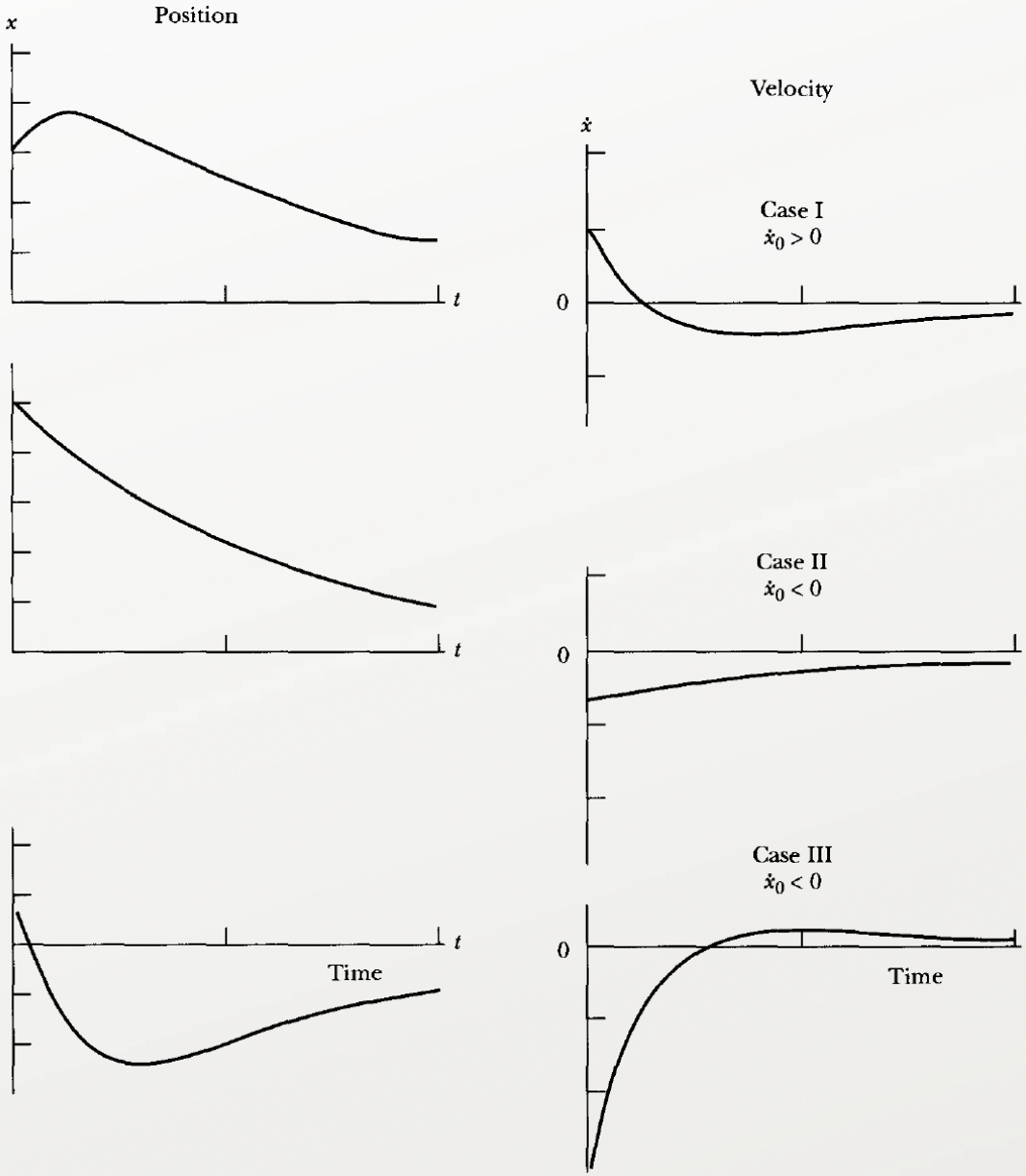
The Phase Diagram of Overdamped Motion

The phase paths for overdamped motion are shown for several initial values of $(x, \dot{x})$. We examined more closely the paths labeled I, II, and III in the previous slide.



The Velocity-Position Behavior of Overdamped Motion

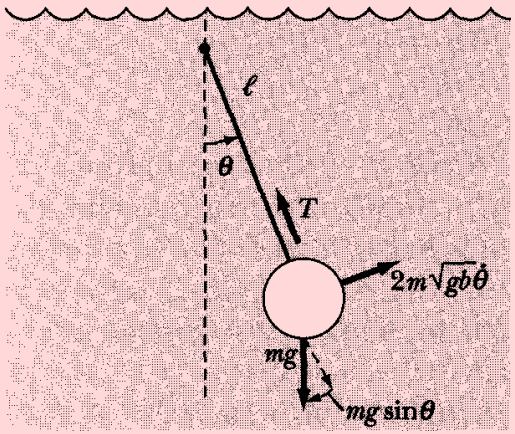
The position and velocity as functions of time for the three phase paths labeled I, II, and III shown in the previous figure.



Problem 8. (Total energy of a Damped Harmonic Oscillator)
Consider a pendulum of length l and a bob of mass mat its end moving through oil with θ decreasing. The massive bob undergoes small oscillations, but the oil retards the bob's motion with a resistive force proportional to the speed with $F_{res} = 2m\sqrt{\frac{g}{l}}(l\dot{\theta})$.

The bob is initially pulled back at $t = 0$ with $\theta = \alpha$ and $\dot{\theta} = 0$.
Find the angular displacement θ and velocity $\dot{\theta}$ as a function of time.

Sketch the phase diagram for $\sqrt{\frac{g}{l}} = 10\text{s}^{-1}$ and $\alpha = 10^{-2}\text{rad}$.



Driven and damped Oscillation
The simplest case of driven oscillation is that in which an external driving force varying harmonically with time is applied to the oscillator. The total force on the particle is then,

$$F = -kx - b\dot{x} + F_0\cos(\omega t)$$

where we consider a linear restoring force and a viscous damping force in addition to the driving force. The equation of motion becomes,

$$\ddot{x} + b\dot{x} + kx = F_0\cos(\omega \text{t})$$

or,

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = A\cos(\omega \text{t})$$

where $A = F_0/m$ and ω is the angular frequency of the driving force. The solution of the equation consists of two parts, a complementary function $x_c(t)$, which is the solution of the equation with the right-hand side set equal to zero, and a particular solution $x_p(t)$, which reproduces the right-hand side.

Problem 9. (Driven-Damped Oscillator) Show that a driven-damped system governed by the above equation has a motion described by $x(t) = x_c(t) + x_p(t)$ where the **complementary solution** $x_c(t)$ is the original solution to the damped oscillator seen before and the **particular solution** $x_p(t) = \frac{A}{\sqrt{(\omega_0^2-\omega^2)+4\omega^2\beta^2}}\cos(\omega t - \delta)$ where $\delta = \tan^{-1}\left(\frac{2\omega\beta}{\omega_0^2-\omega^2}\right)$.

Driven and damped Oscillation

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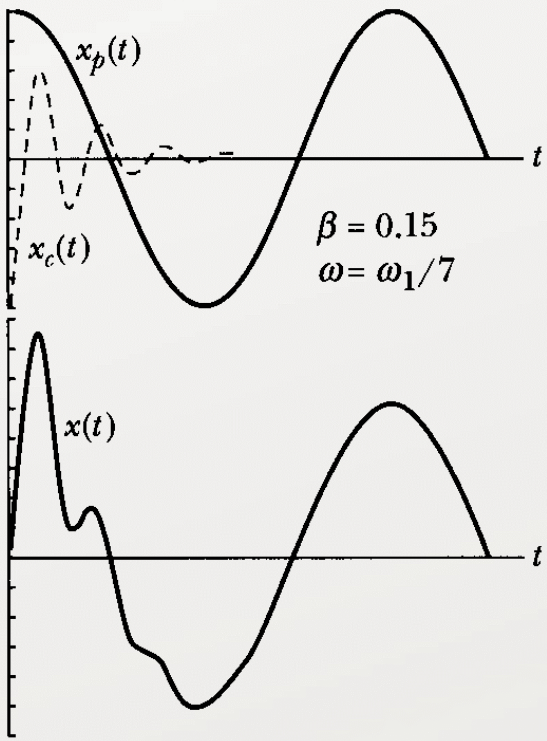
An example of oscillatory driven-damped motion is shown below with the complementary and particular solutions shown in the top plot and the resulting motion is shown in the bottom plot.



Resonant Phenomena

Recall the equation for the amplitude of the particular solution of a driven motion:

$$D = \frac{A}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\omega^2\beta^2}}$$

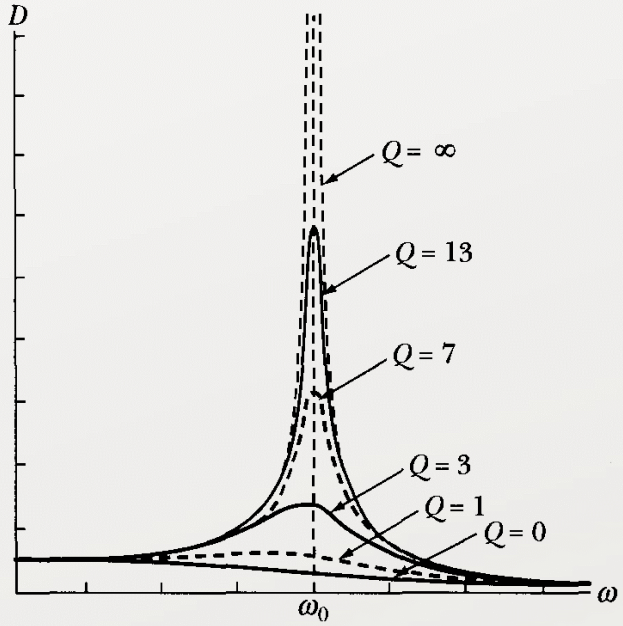
The largest amplitude occurs at the **amplitude resonance frequency**, which is determined via,

$$\left.\frac{dD}{d\omega}\right|_{\omega=\omega_R} = 0$$

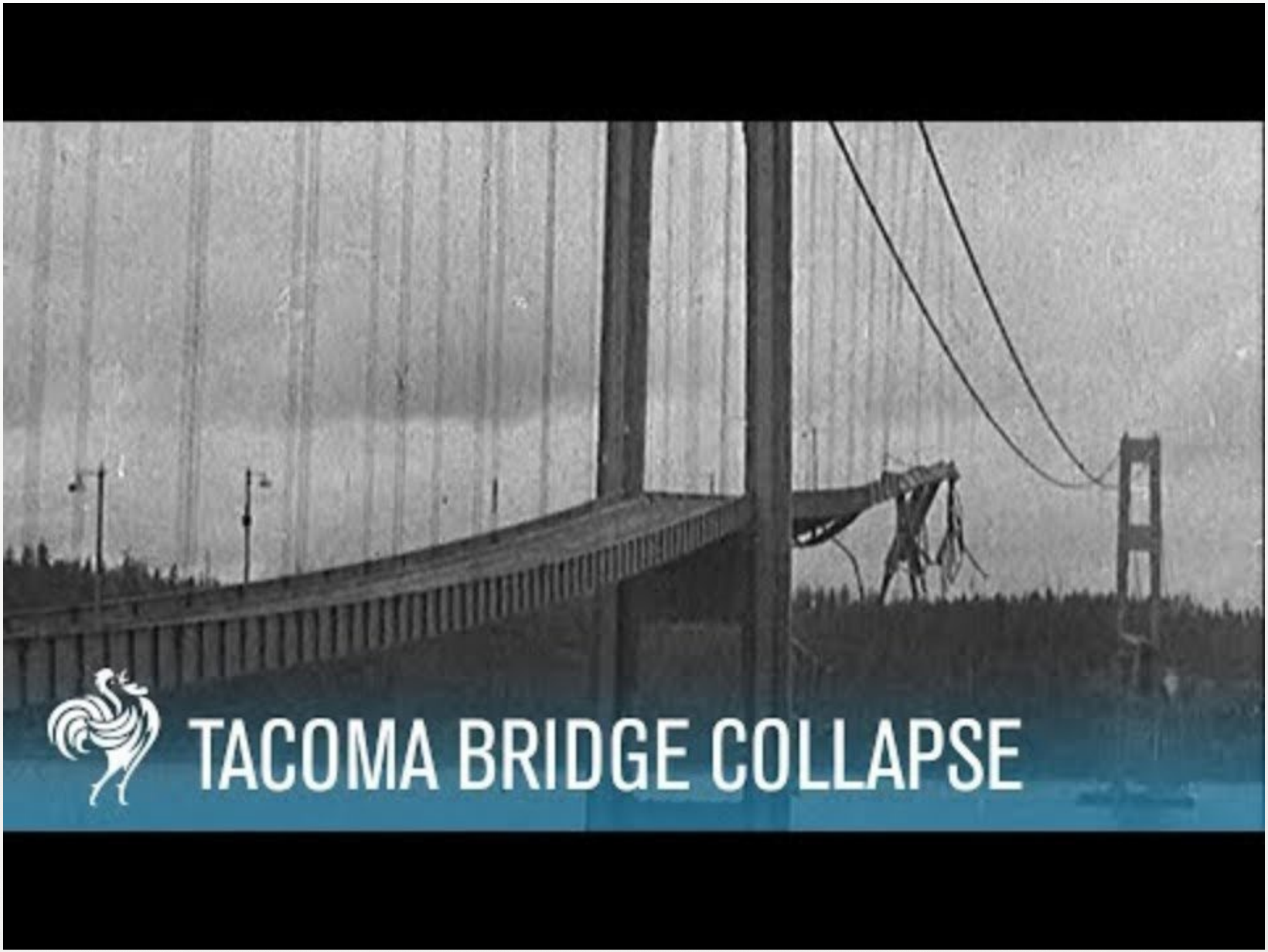
Which yields, $\omega_R = \sqrt{\omega_0^2 - 2\beta^2}$ as the **amplitude resonant frequency**. For comparison,

1. Free oscillations with no damping yields, $\omega_0^2 = \frac{k}{m}$
2. Free oscillations with damping yields $\omega_1^2 = \omega_0^2 - \beta^2$
3. Driven oscillations with damping yields, $\omega_R^2 = \omega_0^2 - 2\beta^2$

In all of the above equations, the condition $\omega_0 > \omega_1 > \omega_R$ holds. It is also customary to define a **Quality Factor** to describe the degree of damping in an oscillatory system as $Q = \frac{\omega_R}{2\beta}$.



Resonant Phenomena



Resonant Phenomena

Problem 10. (The kinetic energy resonance frequency) Show that unlike the amplitude resonance frequency of a damped-driven oscillator, its kinetic resonance frequency is at $\omega_E = \omega_0$ where ω_0 is the natural frequency of the system for undamped oscillations. How about the potential energy resonance frequency? How could this difference be reconsolidated?

Physical Systems

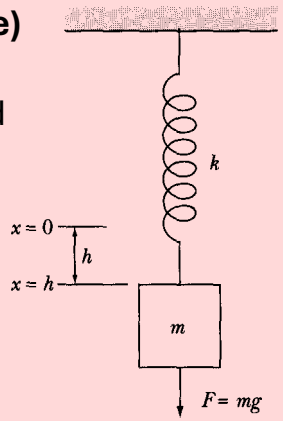
Systems with oscillatory motion go well beyond mechanical pendulums and similar systems. Electrical circuits are the most noted examples of nonmechanical oscillations. Indeed, because of its great practical importance, the electrical example has been so thoroughly investigated that the situation is frequently reversed, and mechanical vibrations are analyzed in terms of the "equivalent electrical circuit."

The following table illustrates the similarity of mechanical and electrical quantities.

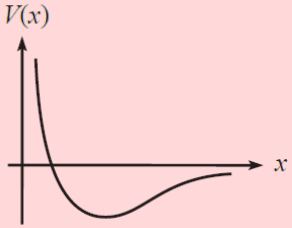
Mechanical		Electrical	
x	Displacement	q	Charge
$\dot{x}$	Velocity	$\dot{q} = I$	Current
m	Mass	L	Inductance
b	Damping resistance	R	Resistance
$1/k$	Mechanical compliance	C	Capacitance
F	Amplitude of impressed force	$\mathcal{E}$	Amplitude of impressed emf

Physical Systems

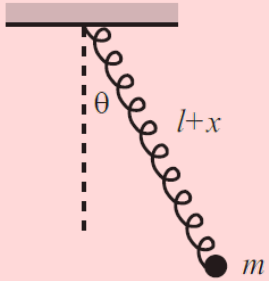
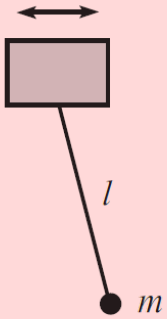
Problem 11. (Electrical circuit equivalence)
Find the equivalent electrical circuit for the hanging mass-spring shown in the figure and determine the time dependence of the charge q in the system.



Problem 12. (Small oscillations frequency)
A particle moves under the influence of the potential $V(x) = \frac{A}{x^2} - \frac{B}{x}$, where $A, B > 0$.
Find the frequency of small oscillations around the equilibrium point.
This potential is relevant to planetary motions.



Problem 13. (Spring pendulum) Consider a pendulum made of a spring with a mass m on the end.
The spring is arranged to lie in a straight line (which we can arrange by, say, wrapping the spring around a rigid massless rod).
The equilibrium length of the spring is l.
Let the spring have length $l + x(t)$, and let its angle with the vertical be $\theta(t)$.
Assuming that the motion takes place in a vertical plane, find the equations of motion for x and θ.



Problem 14. (Pendulum with an oscillating support)
A pendulum consists of a mass m and a massless stick of length l.
The pendulum support oscillates horizontally with a position given by $x(t) = A\cos(\omega t)$; What is the general solution for the angle of the pendulum as a function of time?